student hours Dec. 6

- No student hours after class
- Tues. 4:30pm Wed. 5:45pm MLC

Final exam Thurs. Dec. 12 10:30am-12:30pm SPARK 333

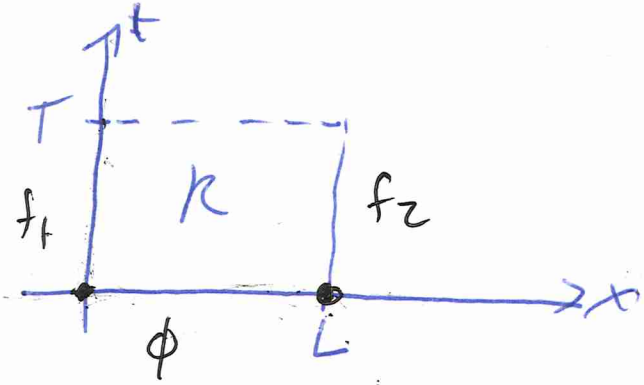
- 4 single sided pp. for notes (or 2 double sided)
- Review topics will be posted today.

Properties of 1D heat eq'n: $u_t - u_{xx} = 0$

① Max-Min Thm

~~from~~

For rectangle R, $T>0$ min/max occurs $\sqcup$



(*) <u>IBVP</u> Find $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$

$$u_t - u_{xx} = 0 \quad 0<x<L,\ t>0$$

$$u(x,0) = \phi(x) \quad 0 \le x \le L$$

Dirichlet B.C.
$$\begin{cases} u(0,t) = f_1(t) & t \ge 0 \\ u(L,t) = f_2(t) & t \ge 0 \end{cases}$$

② smoothness of sol'n to IBVP

$C^\infty((0,L), t>0)$

③ IBVP with Dirichlet B.C. is well-posed

ϕ, compatibility condition

$$\phi(0) = f_1(0)$$

$$\phi(L) = f_2(0)$$

$$f_1, \phi, \phi_2 \in C^0$$

④ IVP with Neumann B.C.

BC, $\frac{\partial u}{\partial x} = \cancel{g_0(x)}$ $\quad \frac{\partial u}{\partial x}(x=L) = \cancel{g_2(t)}\ g_2(t)$

$x=0$: $\quad = g_1(t)$

$\frac{\partial u}{\partial n}$

- We got one sol'n in our example.
- ~~x~~ replacing u by $u+c$

is a sol'n $\quad$ is not a sol'n

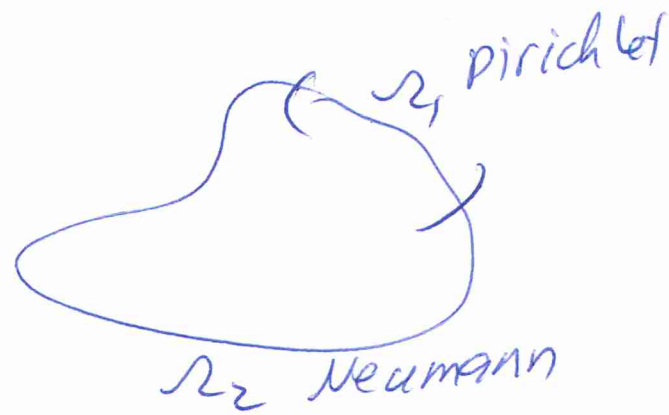
$(u+c)(x,0) = u(x,0) + c = \phi(x) + c$

This is well-posed.

We don't need $\int_{\partial\Omega} g_1(t)\,dx = 0$

Need $g_1, g_2 \in C^0(t>0)$

(5) Mix of Dirichlet + Neumann also well-posed



Backward Heat Eq'n.

Review: wave eq'n, want sol'n for $t=0$

Replace t with t', $t'=-t$

subst. in.

$$u_{xx}-u_{tt}=0 \Rightarrow u_{xx}-u_{t't'}=0$$

solve in $u_{t't'}$ space, replace t' with $-t$.

Heat eq'n. $t'=-t$

$$\cancel{u_t - u_{xx}=0} \quad u_{xx} - u_t = 0 \Rightarrow$$

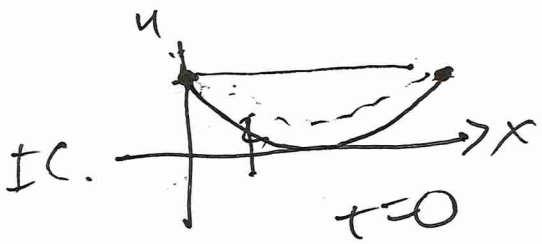
$$u_{xx} + u_{t'} = 0 \quad t>0$$

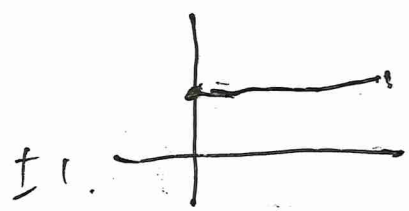
$u(x,0)=\phi(x)$

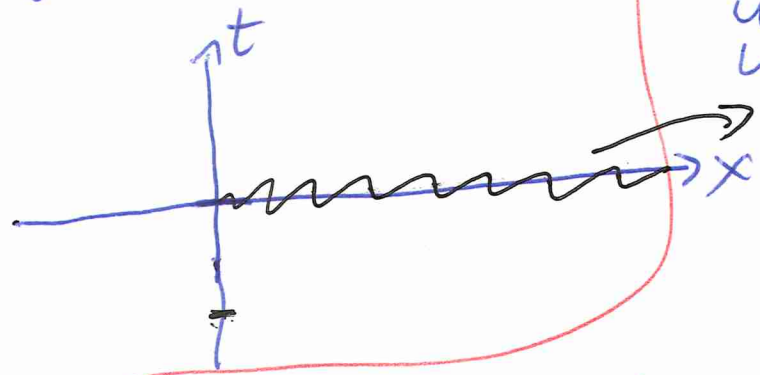
$u(0,t)=f_1(t)$

no temp.

$u(L,t)=f_2(t)$







∞-many sol'ns

Ill-posed

ex) Black-Scholes model Not responsible for

$$\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV + \frac{\partial V}{\partial t} = 0$$

dep. variable : V r, σ are const.

indep. variable S, t

Principle Part $\underbrace{\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + \frac{\partial V}{\partial t}}_{\text{backward heat eq'n}} = 0$

Prescribe $V(t=T)$ solve it for $0 < t < T$

$$\Leftrightarrow \quad \frac{\partial^2 V}{\partial S^2} - \frac{\partial V}{\partial t'} = 0$$

$$V(t'=0) = \ldots$$

Review M of characteristics

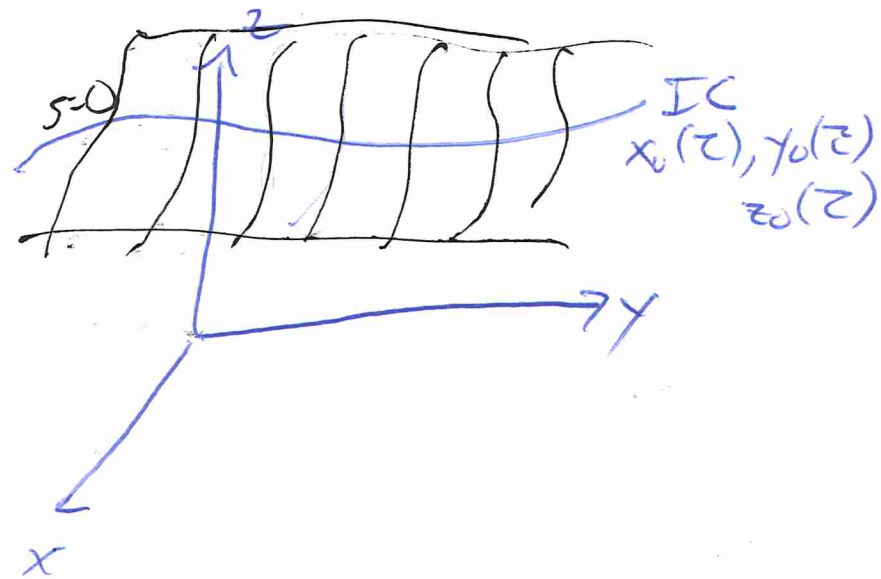
$$P(x,y,z)\, z_x + Q(x,y,z)\, z_y = R(x,y,z)$$

I.C. $(x_0(\tau), y_0(\tau), z_0(\tau))$ $\qquad P, Q, R \in C$

Build a surface from IC using charact. curves $(x(s), y(s), z(s))$ $s=0$ being on I.C.

ex) $x=0,\ z=y^2$

$x=0,\ y=\tau,\ z=\tau^2$



The sol'n exists + is unique if

$$\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0' & y_0' \end{vmatrix} \neq 0$$

↑ only a f'n of τ

The charact. are found by solving

$$\left.\begin{array}{lll} x'(s) = P & y'(s) = Q & z'(s) = R \\ x(s=0) = x_0(\tau) & y(s=0) = y_0(\tau) & z(s=0) = z_0(\tau) \end{array}\right\} \text{system of 3 ODE's \& 3 unknowns}$$

$\Rightarrow z(s,\tau)$

Then ~~use~~ $(s,\tau) \leftrightarrow (x,y) \Rightarrow z(x,y)$

<u>from our notes</u>

ex) $x z_x + (x+y) z_y = z+1 \qquad z(x,0) = x^2$

$P = x \qquad Q = x+y \qquad R = z+1 \qquad x_0(\tau) = \tau$

$y_0(\tau) = 0$

$z_0(\tau) = \tau^2$

exist + uniq?

$$\begin{vmatrix} P|_\Gamma & Q|_\Gamma \\ x_0' & y_0' \end{vmatrix} = \begin{vmatrix} \tau & \tau \\ 1 & 0 \end{vmatrix} = -\tau \neq 0$$

We might have problems at $(0,0,0)$

$$\frac{dx}{ds} = x \qquad \frac{dy}{ds} = x + y \qquad \frac{dz}{ds} = z + 1$$

$$x(s=0) = \tau \qquad y(s=0) = 0 \qquad z(s=0) = \tau^2$$

<u>See notes</u>